# Industrial Internship Report on

# ”Quiz Game”

## Prepared by


## Padarti Dayana Bharathi



*Executive Summary*

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on developing a real-world project within six weeks. My project was **"Quiz Game Using Python and Flask."**

The Quiz Game is a web-based application developed using **Python, Flask, HTML, and CSS**. It presents multiple-choice questions to users, evaluates their answers, calculates the final score, and displays the result instantly. The project helped me understand Flask routing, template rendering, backend logic, and frontend integration.

This internship gave me practical experience in Python web development and improved my problemsolving, debugging, and project development skills. It was a valuable learning experience that strengthened both my technical knowledge and confidence.

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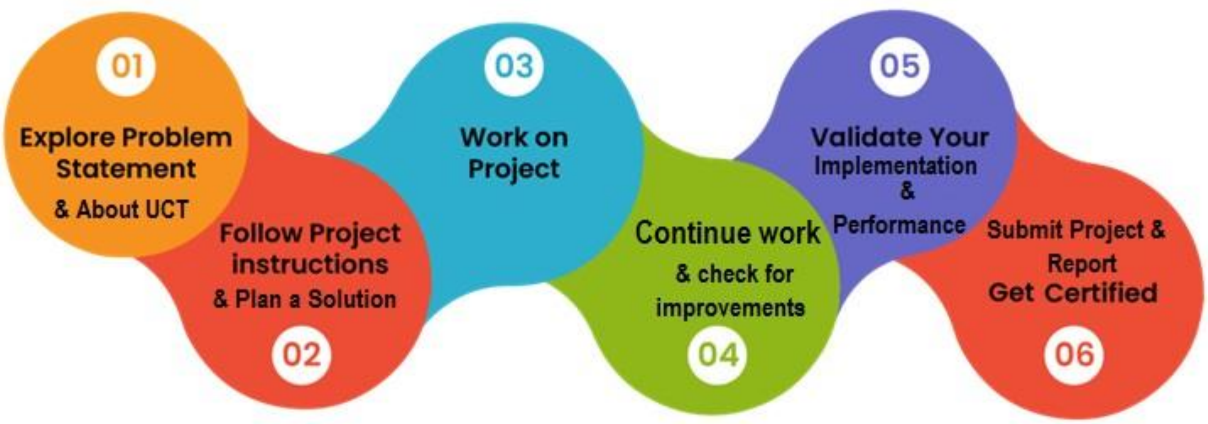
# 1 Preface

This internship provided by upskill Campus and The IoT Academy gave me an opportunity to gain practical experience by developing a real-world project. The project assigned to me was "Quiz Game Using Python and Flask." During this internship, I learned Python web development, Flask, HTML, and CSS while building a quiz application. The internship improved my technical knowledge, problem-solving skills, and confidence in developing web applications.

**The internship was conducted over six weeks with regular learning activities, project development, and report preparation. This structured approach helped me complete the project successfully.**




I sincerely thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT), my college, and my faculty members for their continuous guidance and support. I encourage my juniors and peers to participate in such internships because they provide practical experience and improve technical skills.

# 2 Introduction

## 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

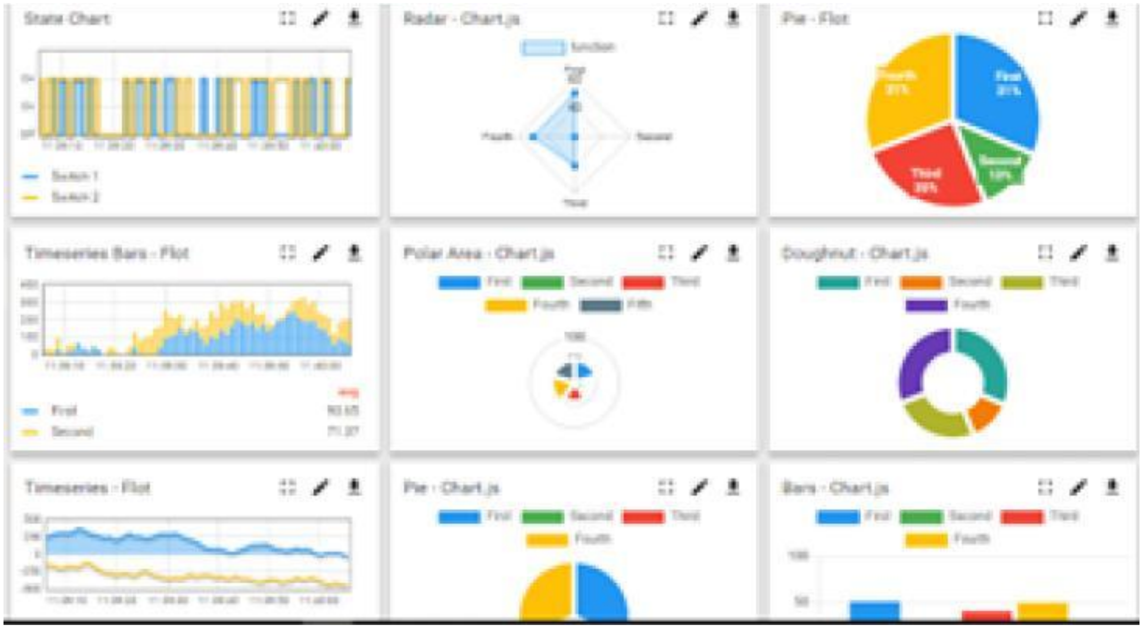

## i. UCT IoT Platform ( Insight )

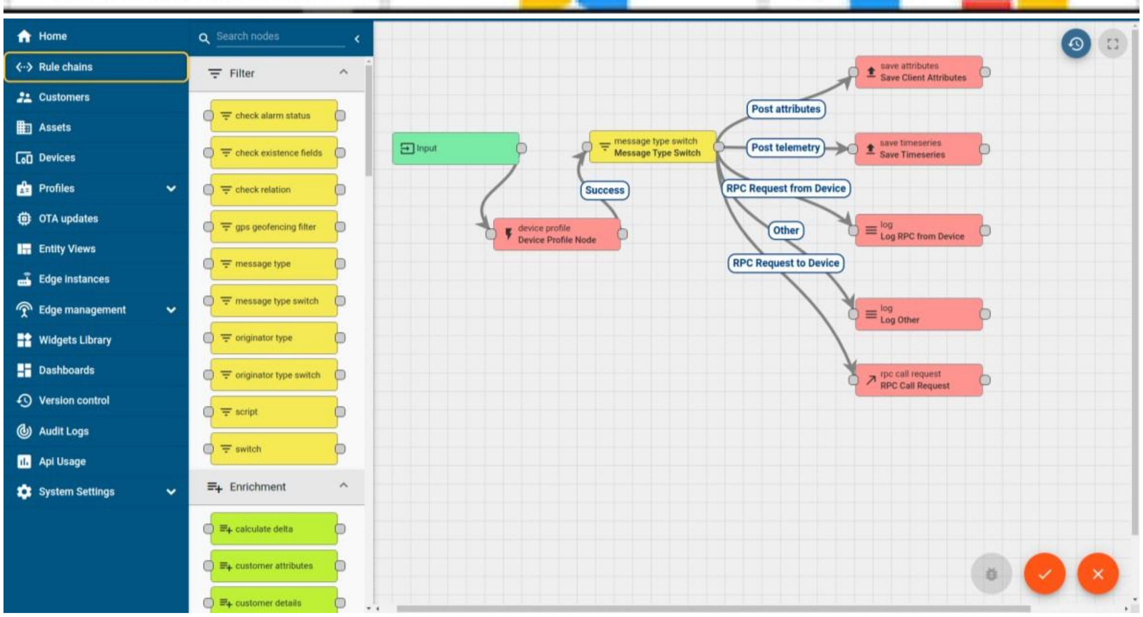
**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine




## ii. Smart Factory Platform ( FACTORY WATCH )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.


| Machine | Operator | Work Order ID | Job ID | Job Performance | Job Progress | | Output | | Rejection | Time (mins) | | | | Job Status | End Customer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Start Time | End Time | Planned | Actual | | Setup | Pred | Downtime | Idle | | |
| CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i |
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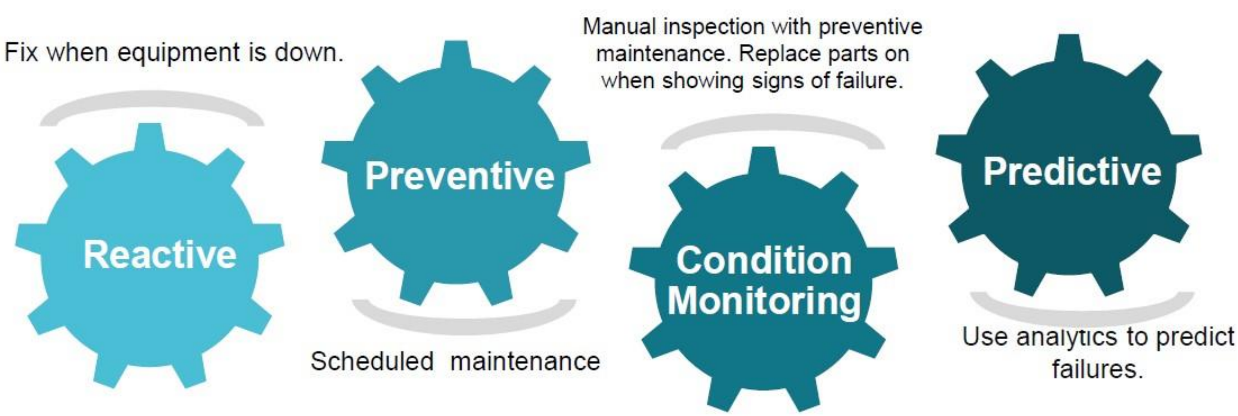
### iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

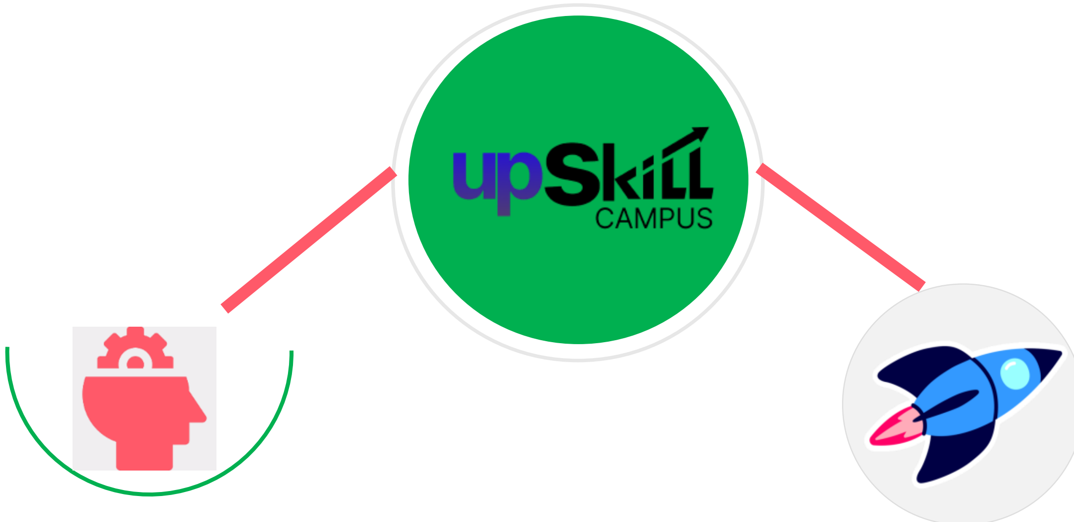
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.




## 2.2 About upskill Campus (USC)

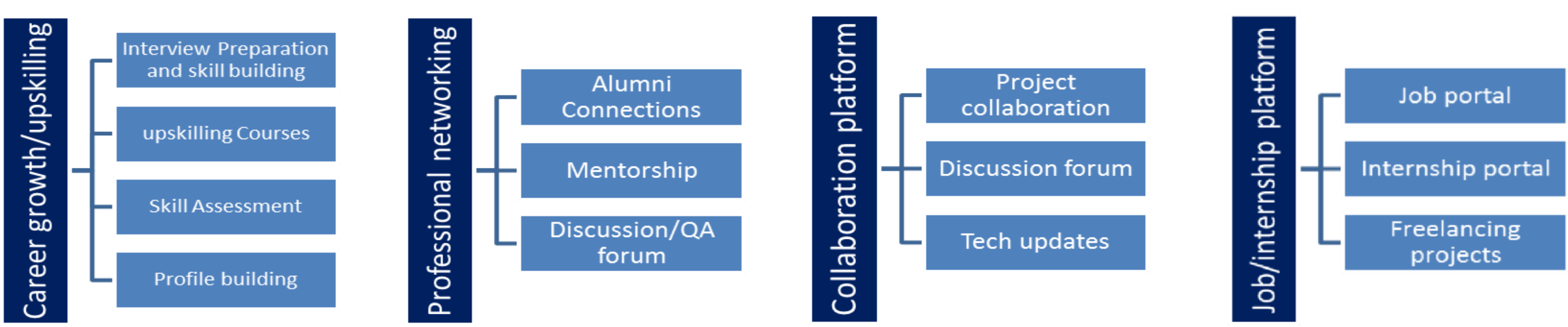
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way



Seeing need of upskilling in self upSkill Campus aiming paced manner along-with
to upskill 1 million additional support services e.g. learners in next 5 year
Internship, projects, interaction with
Industry experts, Career
growth Services

https://www.upskillcampus.com/



### 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

### 2.4 Objectives of this Internship program

The main objectives of this project are:

- To develop a web-based Quiz Game using Python and Flask.
- To provide an interactive platform for answering multiple-choice questions.
- To calculate and display the user's score accurately.
- To improve programming and web development skills using Python, HTML, and CSS.
- To gain practical experience in developing a real-world application during the internship.

### 2.5 Reference

[1] Python Official Documentation – https://docs.python.org/

[2] Flask Official Documentation – https://flask.palletsprojects.com/

[3] HTML & CSS – https://developer.mozilla.org/

### 2.6 Glossary

| Terms | Acronym |
|---|---|
| Python | Programming Language |
| Flask | Python Web Framework |
| HTML | HyperText Markup Language |
| CSS | Cascading Style Sheets |
| UI | User Interface |

## 3 Problem Statement

The objective of this project is to develop a **web-based Quiz Game** using **Python and Flask**. The application should present multiple-choice questions to users, evaluate their answers, calculate the final score, and display the result. The system should provide a simple, interactive, and user-friendly interface while helping users test their knowledge efficiently.

# 4 Existing and Proposed solution

**Existing Solution**

Traditional quiz systems are mostly paper-based or simple online forms. They do not provide an interactive interface or instant score calculation. Users often have to wait for manual evaluation of their answers.

**Proposed Solution**

The proposed solution is a **Quiz Game Using Python and Flask**. The application presents multiple-choice questions through a web interface, evaluates the user's answers automatically, calculates the final score, and displays the result instantly. The project is designed with HTML and CSS to provide a simple and userfriendly experience.

**Value Addition**

- Provides instant score calculation.
- Easy-to-use and interactive interface.
- Saves time by evaluating answers automatically.
- Can be extended by adding more quiz questions in the future.

## 4.1 Code submission (Github link)

https://github.com/padartidayanabharathi/upskillCampus/blob/main/QuizGame.py

## 4.2 Report submission (Github link) :

https://github.com/padartidayanabharathi/upskillCampus/blob/main/QuizGame_DayanaBharathi_USC_UCT.pdf

# 5 Proposed Design/ Model

The Quiz Game is designed as a web-based application using **Python, Flask, HTML, and CSS**. The application displays quiz questions to the user, accepts the selected answers, evaluates the responses, calculates the final score, and displays the result on a separate page. The design is simple, user-friendly, and easy to maintain.

## 5.1 High Level Diagram (if applicable)

Start

↓

Open Quiz Game

↓

Display Questions

↓

User Selects Answers

↓

Submit Quiz

↓

Calculate Score

↓

Display Result

↓

End

**Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM**

### 5.2 Low Level Diagram (if applicable)

User

↓

index.html

↓

Flask (app.py)

↓

questions.py

↓

result.html

**Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM**

### 5.3 Interfaces (if applicable)

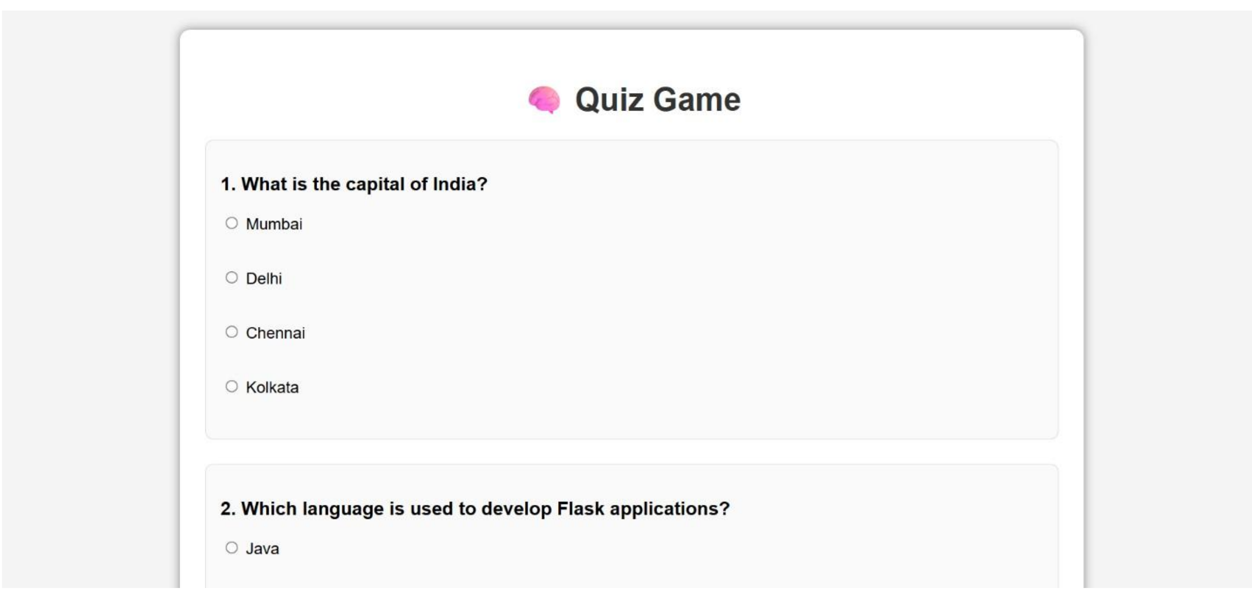



**Figure 3: Quiz Game Home Page**

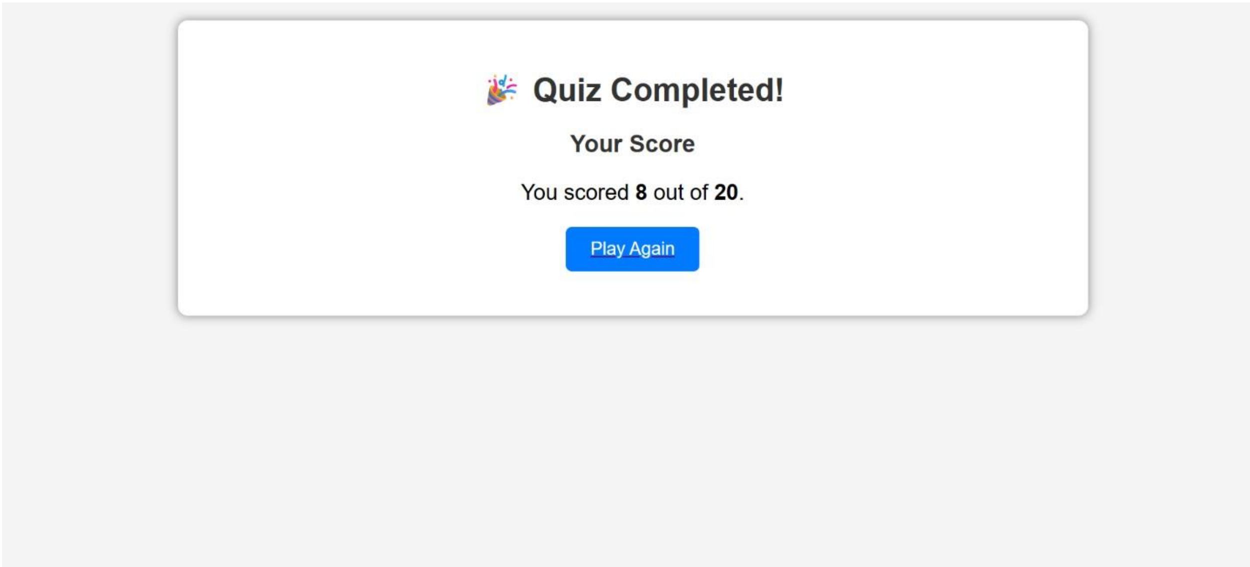



**Figure 4: Quiz Game Result Page**

## 6 Performance Test

The Quiz Game application was successfully tested in the local development environment using the Flask framework. All quiz questions were displayed correctly, user responses were processed successfully, and the final score was calculated accurately.

**Test Results**

- The application started successfully using the Flask server.
- All quiz questions were displayed correctly.
- User answers were validated successfully.
- The final score was calculated and displayed accurately.
- The application worked without any runtime errors during testing.

## 6.1 Test Plan/ Test Cases

The test plan was performed to verify the functionality of the Quiz Game application. Different test cases were executed to ensure that the application works correctly.

**Test Case 1: Application Launch Testing**
The Flask server was started successfully, and the application loaded without errors.

**Test Case 2: Question Display Testing**
The quiz questions and options were displayed correctly to the user.

**Test Case 3: Answer Validation Testing**
User-selected answers were recorded and validated successfully.

**Test Case 4: Score Calculation Testing**
The system calculated the final score accurately based on the submitted answers.

**Test Case 5: Result Display Testing**
The final result was displayed successfully after quiz completion.

The test results confirmed that all major functionalities of the Quiz Game application were working properly.

## 6.2 Test Procedure

The following steps were followed to test the Quiz Game application:

1. The Flask application was executed using the local server.
2. The quiz page was opened in the browser.
3. Questions and answer options were verified.
4. User responses were selected and submitted.
5. The score calculation and final result display were checked.

The application performed successfully during the testing process without any errors.

### 6.3 Performance Outcome

The performance evaluation of the Quiz Game application showed that the system operated efficiently under normal usage conditions.

- The application loaded quiz questions quickly without delays.
- User inputs were processed accurately.
- Score calculation was completed instantly after quiz submission.
- The Flask server handled the application requests successfully.
- The application provided a smooth and user-friendly experience during testing.

The overall testing results confirmed that the Quiz Game application met the required functional and performance requirements.

# 7 My learnings

During the development of the Quiz Game application, I gained practical knowledge of web application development using the Flask framework. I learned how to create a user-friendly interface, handle user inputs, process data, and implement application logic using Python.

The project helped me improve my understanding of Flask routing, HTML, CSS, and Python programming concepts. I also learned the importance of testing, debugging, and validating application performance to ensure a reliable user experience.

## 8 Future work scope

The Quiz Game application can be enhanced with additional features in the future. User authentication can be added to maintain individual user profiles and track quiz history. A database can be integrated to store questions, user scores, and performance records.

The application can also be improved by adding different difficulty levels, categories, timers, leaderboards, and an interactive dashboard. These enhancements can make the system more engaging, scalable, and suitable for real-world educational platforms.